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D-ITG - DISTRIBUTED INTERNET TRAFFIC GENERATOR

www.grid.unina.it/software/ITG

The **Distributed Internet Traffic Generator (D-ITG)** is a platform capable to produce traffic (network, transport and application layer) and accurately replicate appropriate stochastic processes for both IDT (*Inter Departure Time*) and PS (*Packet Size*) i.i.d. random variables (*exponential, uniform, cauchy, normal, pareto, ...*).

D-ITG is currently available on **Linux, Windows** and **Linux Familiar** platform. It presents both a **multithread** and a **multitask** implementation. The supported protocols are: **TCP, UDP, ICMP, DNS, Telnet, VoIP** (G.711, G.723, G.729, Voice Activity Detection, Compressed RTP). The provided stochastic processes both in the case of PS and IDT are **Constant, Uniformly distributed, Exponentially distributed, Pareto distributed, Cauchy distributed, Normal distributed, Poisson distributed, Gamma distributed**. D-ITG provides setting of generation seed: this option gives the possibility to repeat different experiments *using the same seed*. D-ITG can perform both *one-way-delay* (OWD) measurement and *round-trip-time* (RTT) measurement, *packet loss* evaluation, *jitter* and *throughput* measurement. Another innovative feature of D-ITG is the possibility to store information both on the receiver and on the sender.

Additionally, D-ITG enables the sender and the receiver to delegate the logging operation to a *remote log server*. This option is useful when the receiver has limited storage capacity - e.g. PDAs, Palmtops, etc. - and when log information must be analyzed “on-the-fly”, for example, in case the sender is asked to adapt the transmission rate based on channel congestion and receiver capacity. D-ITG permits the setting of *TOS* (DS) and *TTL* packet field. Both experiment *duration* and *delay* (with respect to initial time of the experiment) can be set. The communication between sender and receiver is done by using a separated *signaling channel* that implements a protocol for the configuration of the experiment (*Traffic Specification Protocol*). Furthermore, the sender can be remotely controlled by using *ITGapi*. This means that the D-ITG sender can be launched in daemon mode and it waits for command (the generation of traffic flows is remotely controlled). By using this feature it is possible to test traffic engineering algorithms in a real network. D-ITG is able to reach high (receiver and sender) data rate. More precisely, in a local environment (sender and receiver over the same Linux platform) the maximum data rate is equal to **511 Mbps** both at sender and receiver side; in a distributed environment (sender and receiver over two different Linux platforms) the maximum data rate is equal to **612 Mbps** at sender side and 611 Mbps at receiver side.

D-ITG is currently downloadable and freely available at www.grid.unina.it/software/ITG.